AI-BASED POTHOLE DETECTION AND ROAD CONDITION MONITORING SYSTEM

Inspection Report

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Source Type: Image

Executive Summary

This report presents the results of automated road inspection using an AI-Based Pothole Detection and Road Condition Monitoring System built on YOLOv8.

Source Type: **Image**
Total Potholes Detected: **4**
Road Condition Score: **68/100**
Overall Road Condition: **GOOD**

Detection Results

Total Potholes Detected: 4

Severity Breakdown
Low Severity : 1
Medium Severity : 3
High Severity : 0

Road Condition Assessment

Road Score: 68/100

Road Condition: GOOD

Score Interpretation:
80-100 : Excellent
60-79 : Good
40-59 : Moderate
20-39 : Poor
0-19 : Critical

Maintenance Recommendations

• Minor road damage detected.
• Repair medium severity potholes.
• Schedule periodic inspections.
• Continue monitoring affected sections.

Risk Assessment

Potential impacts of detected potholes include:

• Tire damage
• Suspension damage
• Increased accident risk
• Reduced ride comfort
• Increased vehicle maintenance costs
• Traffic congestion due to road deterioration

Technical Information

Detection Model: YOLOv8
Framework: Ultralytics
Input Type: Image
Analysis Type: Automated Road Inspection
Output: Pothole Detection, Severity Classification and Road Condition Assessment

Conclusion

The AI-Based Pothole Detection System successfully analyzed the provided image and detected **4** potholes.

Based on severity classification and road condition evaluation, the road received a score of **68/100** and was classified as **GOOD**.

Appropriate maintenance actions are recommended to improve road quality, safety and driving comfort.

This report was generated automatically by the AI-Based Pothole Detection and Road Condition Monitoring System.